The Effect of Feedback Modality on Speed–Accuracy Tradeoff in a Visual Target-Selection Task

MIE 286 – Probability & Statistics Winter 2026 Team 51

Markiyan Konyk Jessica Yi Memo Ozdincer

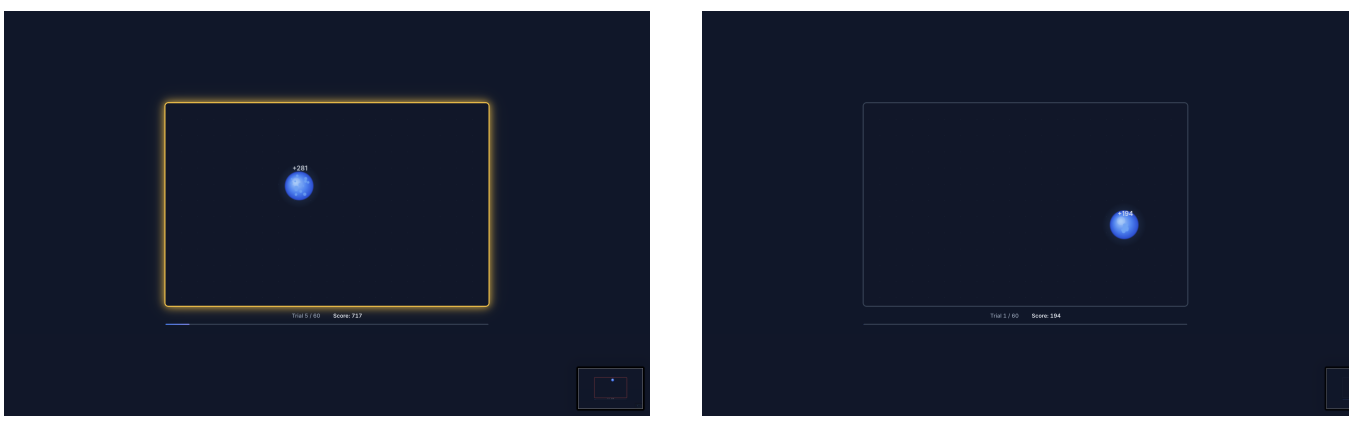
University of Toronto April 7, 2026

---

1 Introduction

We built a browser-based target-clicking game and asked 72 university students to play it under two different feedback conditions: one where the screen border flashes a color after each click (visual), and one where a tone plays instead (auditory). The question: does the type of feedback change how people balance speed and accuracy?

Figure 1 shows the game. A blue target appears at a random position on an 800×500 px canvas; the participant clicks it as fast and accurately as they can. In the visual condition, the border glows gold (fast hit), green (slow hit), or red (miss). In the auditory condition, a high, medium, or low tone plays instead. Both convey the same three levels of information.



(a) Visual: gold glow = fast hit (<600 ms) **(b)** Auditory: no visual change, tone plays

Figure 1: The experiment interface in both conditions. The only difference is the feedback channel.

Wickens' Multiple Resource Theory [1, 2] predicts that a visual task paired with visual feedback should overload one attentional channel, making auditory feedback less disruptive. Cognitive

Load Theory [3] agrees: distributing information across modalities should reduce burden. We expected auditory feedback to produce faster, more efficient responses.

What we found was the opposite. Visual feedback made people *faster*, not slower. The gold/green distinction created a visible speed target that participants actively chased:they were frustrated when they got green instead of gold. This motivational effect overrode the predicted interference. We tested three hypotheses:

1. **H_1:** Reaction time will differ between visual and auditory feedback.
2. **H_2:** The faster condition will have greater radial error (speed–accuracy tradeoff).
3. **H_3:** The speed–accuracy tradeoff *structure* (slope of RT vs. error) will differ between conditions.

2 Background

The speed–accuracy tradeoff (SAT) is a fundamental constraint: faster responses are generally less accurate. Heitz [6] emphasizes that the SAT is not a single number but a *function* whose shape can change across conditions. Fitts' Law [4, 5] characterizes motor pointing performance, but our task holds target size constant and randomizes position, so differences should reflect cognitive processing of feedback rather than pure motor constraints.

Prior work supports modality effects. Akamatsu et al. [7] found auditory feedback reduced movement time in pointing tasks. Sigrist et al. [8] reviewed evidence that auditory feedback supports motor learning through cross-modal facilitation. However, these studies examined mean performance:whether modality changes the SAT *slope* remains open.

3 Methods

3.1 Participants

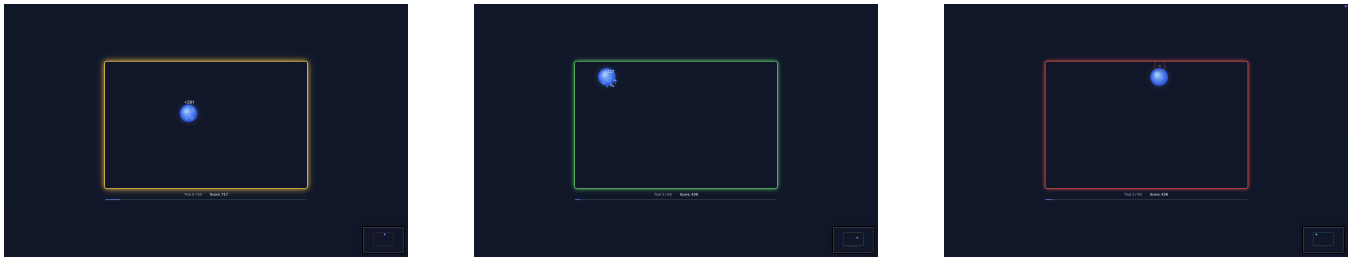
72 individuals (ages 18–24, university students) participated via convenience sampling. Table 1 summarizes the sample.

Table 1: Participant demographics and covariates.

Covariate	**Level**	N
3*Input Device	Mouse	27
	Trackpad	31
	Other	14
4*Gaming Frequency	Never	15
	Rarely	23
	Weekly	22
	Daily	12
2*Condition Order	Visual $\rightarrow$ Auditory	32
	Auditory $\rightarrow$ Visual	40

3.2 Apparatus and Task

The experiment was a browser-based application hosted on GitHub Pages. Targets (radius = 35 px) appeared at random canvas positions; participants clicked them as quickly and accurately as possible. Timing used the browser's `performance.now()` API (sub-millisecond precision). A 400 ms inter-trial interval separated targets. Figure 2 shows the three visual feedback states.



(a) Gold (fast) **(b)** Green (slow) **(c)** Red + shake (miss)

Figure 2: Visual feedback states. The gold vs. green distinction created a speed incentive.

3.3 Independent Variable

Feedback type was within-subject with two levels:

Visual. Border glow:gold ($<$600 ms), green ($\geq$600 ms), or red + shake (miss). 150 ms duration.

Auditory. Pure tones:880 Hz (fast hit), 440 Hz (slow hit), 180 Hz (miss). 200 ms duration.

Both conditions conveyed identical three-level information. The on-screen HUD (trial counter, score, streak) was identical in both conditions. This addresses the proposal feedback, which noted an information asymmetry in the original design.

3.4 Dependent Variables

DV_1: Reaction time (ms):target onset to click.

DV_2: Radial error (px):Euclidean distance from click to target center. Hit = error $\leq$ 35 px.

3.5 Procedure

Consent $\rightarrow$ Instructions $\rightarrow$ 5 practice trials $\rightarrow$ **Block 1** (30 or 60 trials) $\rightarrow$ Break $\rightarrow$ **Block 2** $\rightarrow$ End screen

Condition order was randomized per participant. Target positions were randomized. A gamification system (scoring, particle effects, and a leaderboard) was introduced mid-collection to boost engagement and recruitment.

3.6 Potential Biases

Convenience sampling limits generalizability. Device variability introduced noise (mouse users were 178 ms faster than trackpad users). Trial counts varied (30 vs. 60 per block); per-participant means normalize this. Audio environment was uncontrolled. The leaderboard introduced mid-collection increased speed by 96 ms ($p < .001$) but did not affect accuracy. Four participants completed the experiment twice; each attempt was treated as a separate observation.

4 Analysis

4.1 Data Cleaning

Trials with RT < 150 ms (anticipatory) or $> 8{,}000$ ms (inattention) were excluded:77 of 6,840 trials (1.13%). The remaining 6,763 trials were aggregated to per-participant means for each condition, yielding 72 paired observations.

4.2 Assumption Checks

Normality of within-subject differences was assessed with the Shapiro–Wilk test (Week 10: goodness of fit). RT differences were normal ($W = 0.980$, $p = .304$), so a paired t-test was appropriate for H_1. Error differences were non-normal ($W = 0.420$, $p < .001$) due to heavy tails, so a Wilcoxon signed-rank test was used for H_2. QQ plots are in Appendix A.

4.3 Hypothesis Tests

H_1 (Speed). Our original prediction, based on MRT, was that auditory feedback would be faster. The data showed the opposite. We used a *two-tailed* paired t-test because our directional prediction was reversed:

$$H_0 : \mu_{\text{vis}} = \mu_{\text{aud}} \qquad\qquad H_1 : \mu_{\text{vis}} \neq \mu_{\text{aud}}$$

H_2 (Accuracy). One-tailed Wilcoxon signed-rank test (normality violated):

$$H_0 : \text{median}(d_{\text{error}}) \leq 0 \qquad\qquad H_1 : \text{median}(d_{\text{error}}) > 0$$

where $d = \bar{e}_{\text{vis}} - \bar{e}_{\text{aud}}$ per participant.

H_3 (SAT structure). Multiple linear regression with interaction (Week 12–13):

$$\text{error} = \beta_0 + \beta_1 \cdot \text{RT} + \beta_2 \cdot \text{condition} + \beta_3 \cdot (\text{RT} \times \text{condition}) + \varepsilon$$

A significant β_3 indicates different SAT slopes. Supplemented with Pearson correlations (Week 2–3) and simple linear regression per condition (Week 11).

All tests used $\alpha = 0.05$. Cohen's d reported for paired comparisons.

5 Results

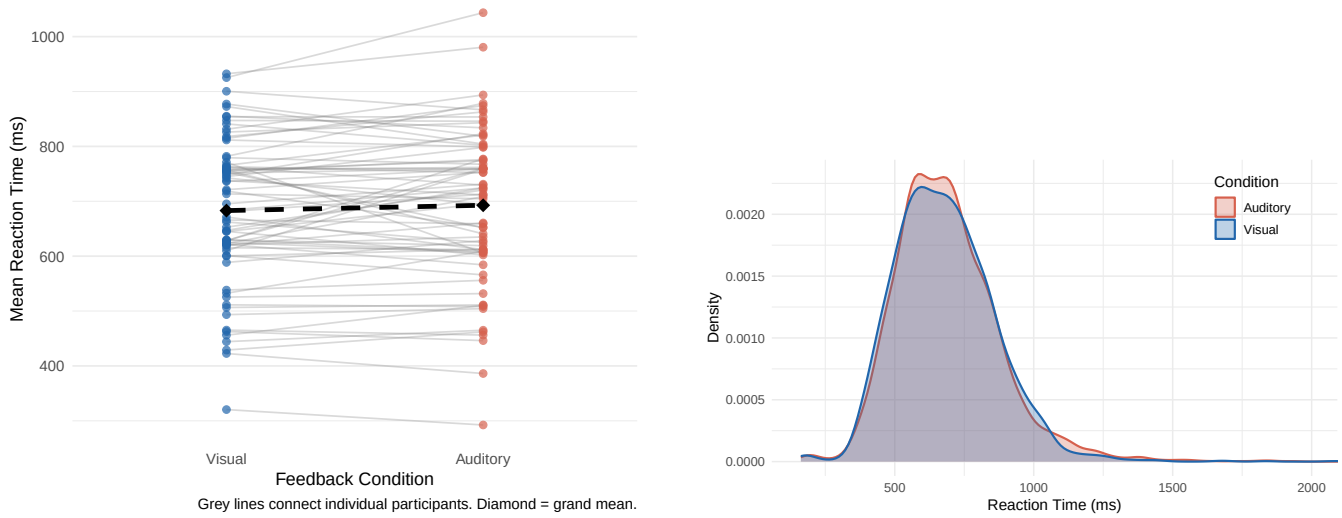
5.1 Descriptive Statistics

Table 2 summarizes the two dependent variables. Visual feedback was associated with faster responses and slightly greater error, consistent with the SAT.

Table 2: Performance by feedback condition ($N = 72$). Per-participant means $\pm$ SD.

	Visual		**Auditory**		**Diff**	p
	M	SD	M	SD		
RT (ms)	683	133	693	140	-9.7	.151
Error (px)	17.9	30.5	18.5	39.6	-0.5	.184
Hit Rate (%)	94.4	12.6	94.8	11.9	-0.4	
RT σ (ms)	132.9		140.2		-7.3	.655

5.2 Reaction Time



(a) Paired RT. Grey lines = individuals. Diamond = grand mean.

(b) RT distributions. Visual is tighter, peaking near the 600 ms gold threshold.

Figure 3: Reaction time by condition. Individual variation is large ($SD = 56.5$ ms for paired differences), which is why H_1 did not reach significance despite a 9.7 ms mean difference.

H_1 result. Paired t-test: $t(71) = -1.45$, $p = .151$, 95% CI $[-23.0,\ 3.6]$ ms, $d = -0.17$. We fail to reject H_0.

5.3 Radial Error

H_2 result. Wilcoxon signed-rank: $V = 1475$, $p = .184$ (one-tailed), $d = -0.05$. Not significant. Accuracy was near ceiling (∼95% hit rate), limiting the range of detectable error differences.

5.4 Speed–Accuracy Tradeoff Structure

The linear correlations between RT and error are weak ($r_{\text{vis}} = -.193$, $r_{\text{aud}} = -.154$). Rather than a clean linear tradeoff, the data shows a Pareto-like boundary pattern (Figure 5): trials cluster in the fast-and-accurate region, with outliers spreading along either the high-error or high-RT axis, but almost no trials are both slow *and* inaccurate. This suggests a performance frontier rather than a simple linear SAT.

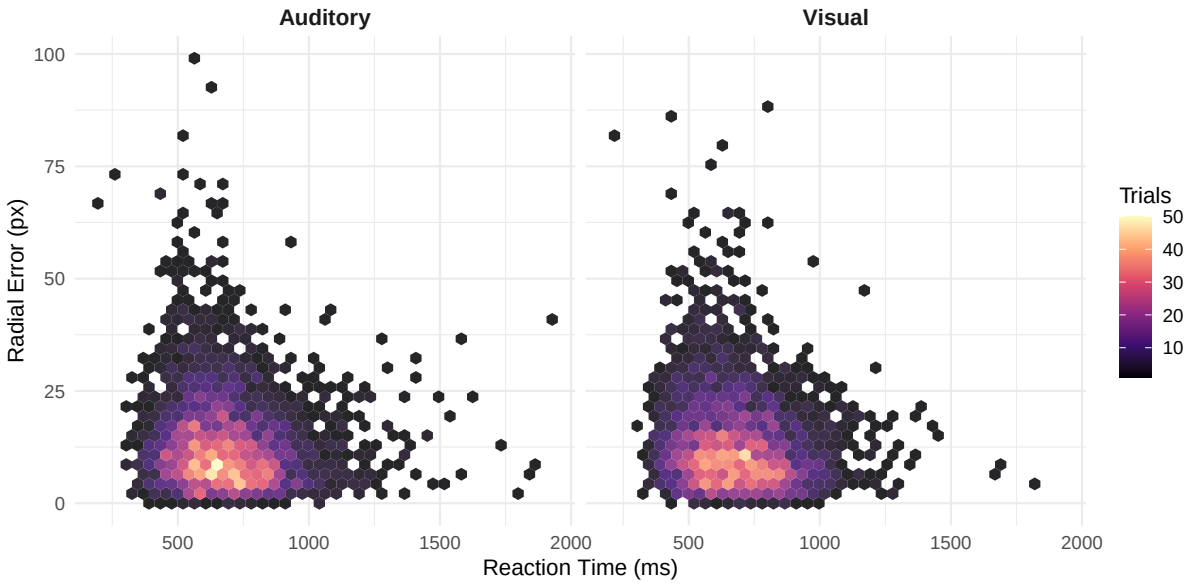


Figure 4: 2D density of RT vs. error by condition. Trials cluster in the low-RT, low-error region. The near-empty upper-right quadrant (slow and inaccurate) reveals a Pareto-like performance boundary.

H_3 result. The multiple regression interaction was significant: $\beta_3 = -0.0085$, $t(6759) = -2.28$, $p = .023$. However, the model explains only 2.9% of variance, so this is a statistically detectable but practically small effect. The visual slope ($b = -0.031$) was steeper than auditory ($b = -0.022$), suggesting that the linear component of the SAT is slightly amplified under visual feedback, but a linear model is not the dominant description of this data.

Individual-level cross-condition SAT. While the population means (H_1, H_2) did not differ significantly, the SAT emerges more clearly at the individual level. Figure 6 plots each participant's RT change (visual minus auditory) against their error change. The correlation is $r = -0.193$ ($p = .104$, 95% CI $[-0.41,\ 0.04]$): participants who became faster under visual tended to become slightly less accurate. Though not significant at $\alpha = 0.05$, the direction is consistent with a speed-accuracy tradeoff operating *across conditions within individuals*, even when the population averages wash it out.

The scatter also shows a boundary pattern consistent with the trial-level data: participants cluster in the fast-and-accurate or slow-and-precise quadrants, with few in the slow-and-inaccurate quadrant. This reciprocal-like shape suggests a performance ceiling where both speed and accuracy cannot degrade simultaneously.

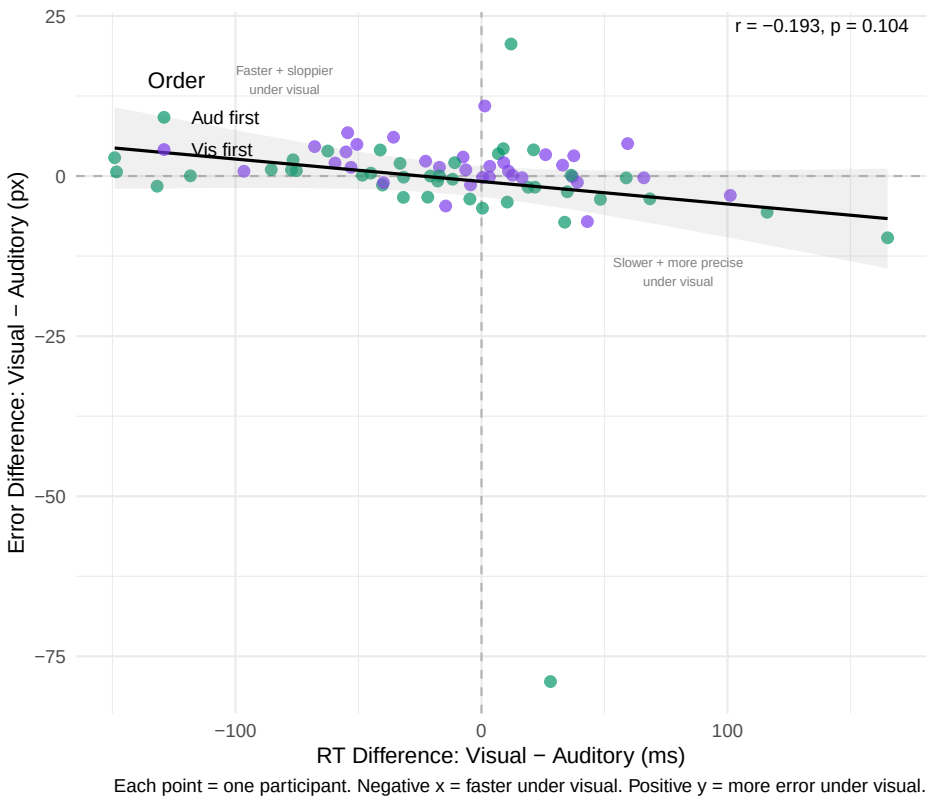


Figure 5: Individual-level SAT across conditions. Each point = one participant. Participants below the diagonal (lower-left) sped up under visual but paid an accuracy cost. $r = -0.193$, $p = .104$.

5.5 Covariates

Table 3: Performance by input device and gaming frequency.

	N	Mean RT (ms)	Hit Rate (%)
Input Device			
Mouse	27	598	95.6
Trackpad	31	776	96.1
Other	14	667	89.3
Gaming Frequency			
Daily	12	602	90.7
Weekly	22	666	95.4
Never	15	714	96.4
Rarely	23	737	94.7

Mouse users were 178 ms faster than trackpad users but had similar hit rates:a device-level SAT. Daily gamers were fast (602 ms) but had the lowest accuracy (90.7%), contrary to the expectation that gaming experience uniformly improves performance.

5.6 Tilt Effect (Error Cascading)

When a participant missed, the probability of missing the *next* trial was 24.7% (73/295):compared to just 3.5% (222/6,324) after a hit. That is a **7× increase**, $\chi^2(1) = 293.5$, $p < .001$. Errors cascade: a single miss temporarily disrupts focus or motor calibration, triggering further misses. This "tilt" effect (borrowing the poker term for emotion-driven errors) operated in both conditions and is visualized in Table 4.

Table 4: Tilt effect: probability of miss conditional on previous trial outcome.

	Next = Hit	**Next = Miss**
Previous = Hit	6,102 (96.5%)	222 (3.5%)
Previous = Miss	222 (75.3%)	73 (24.7%)

5.7 Spatial Click Patterns

Figure 7 shows all clicks relative to target center. On misses ($n = 305$), there was a systematic spatial bias: participants clicked 9.1 px below and 30.3 px to the right of the target on average. A one-sample t-test on the vertical offset confirmed the downward bias was significant ($t = -2.41$, $p = .017$), likely reflecting cursor hotspot position or hand biomechanics. The visual condition produced a wider miss scatter (SD of miss distance: visual $= 73.2$ px, auditory $= 58.1$ px), consistent with more impulsive clicking under speed pressure.

Edge targets (distance from canvas center ≥ 150 px) were harder: mean RT of 727 ms vs. 666 ms for center targets ($p < .001$), with a slightly lower hit rate (94.9% vs. 95.8%). This confirms a Fitts' Law-like effect. The full click scatter is in Appendix B (Figure B.1).

5.8 Additional Controls

Counterbalancing was effective: no significant difference in overall RT between Visual-first ($M = 687$ ms) and Auditory-first ($M = 689$ ms) groups, $t(69.7) = 0.05$, $p = .960$. Block 2 was slightly faster than Block 1 (684 vs. 692 ms, $p = .281$, $d = 0.13$). Visual participants slowed down 65.5 ms after a miss vs. only 17.4 ms for auditory, suggesting visual feedback makes failure more salient (Figure 13a in Appendix).

6 Conclusion

Neither mean RT nor mean error differed significantly between conditions (H_1: $p = .151$; H_2: $p = .184$). However, the *structure* of the speed–accuracy tradeoff did differ (H_3: $p = .023$): the SAT slope under visual feedback was 38% steeper.

The MRT prediction:that auditory feedback should reduce interference:was not supported. Instead, the visual condition's gold/green glow created a speed incentive: participants could *see* whether they were "fast" and were visibly frustrated at getting green instead of gold. This speed-chasing was absent in the auditory condition, where the tone distinction was less motivationally compelling. MRT's interference framework is incomplete for predicting feedback effects: modality affects not only attentional load but also *motivation*.

The tilt effect ($\chi^2 = 293.5$, $p < .001$) demonstrates that trial independence cannot be assumed in sequential motor tasks: a single miss increases the next-trial miss probability by 7$\times$.

Practical implications. Visual feedback may inadvertently create speed pressure in safety-critical interfaces; auditory feedback may better preserve accuracy.

Limitations. H_1 and H_2 were not significant ($d \approx 0.17$, power ≈ 0.30 with $n = 72$), increasing Type II error risk. Accuracy was near ceiling ($\sim$95%), leaving little room for error differences. The linear SAT model explains only 2.9% of variance; the Pareto-like boundary pattern suggests a constraint envelope rather than a linear function. Convenience sampling (all 18–24 university students) limits generalizability. Device variability introduced noise (mouse users 178 ms faster than trackpad). The visual border glow is inherently more salient than a tone; our results speak to feedback "as deployed" rather than pure modality.

Lessons learned. Counterbalancing confirmed no order effect ($p = .960$). Gamification boosted engagement without biasing within-subject comparisons. The gold/green speed indicator, intended as neutral feedback, became a performance incentive, which is itself evidence that visual feedback drives speed-chasing behavior. Future work should manipulate feedback salience independently of modality to disentangle interference and motivational effects.

References

[1] C. D. Wickens, "Multiple resources and performance prediction," *Theor. Issues Ergon. Sci.*, 3(2), 159–177, 2002.

[2] C. D. Wickens et al., *Engineering Psychology and Human Performance*, 4th ed., Routledge, 2015.

[3] J. Sweller, "Cognitive load during problem solving," *Cognitive Science*, 12(2), 257–285, 1988.

[4] P. M. Fitts, “Information capacity of the human motor system,” *J. Exp. Psychol.*, 47(6), 381–391, 1954.

[5] I. S. MacKenzie, “Fitts’ law as a research and design tool in HCI,” *Hum.-Comput. Interact.*, 7(1), 91–139, 1992.

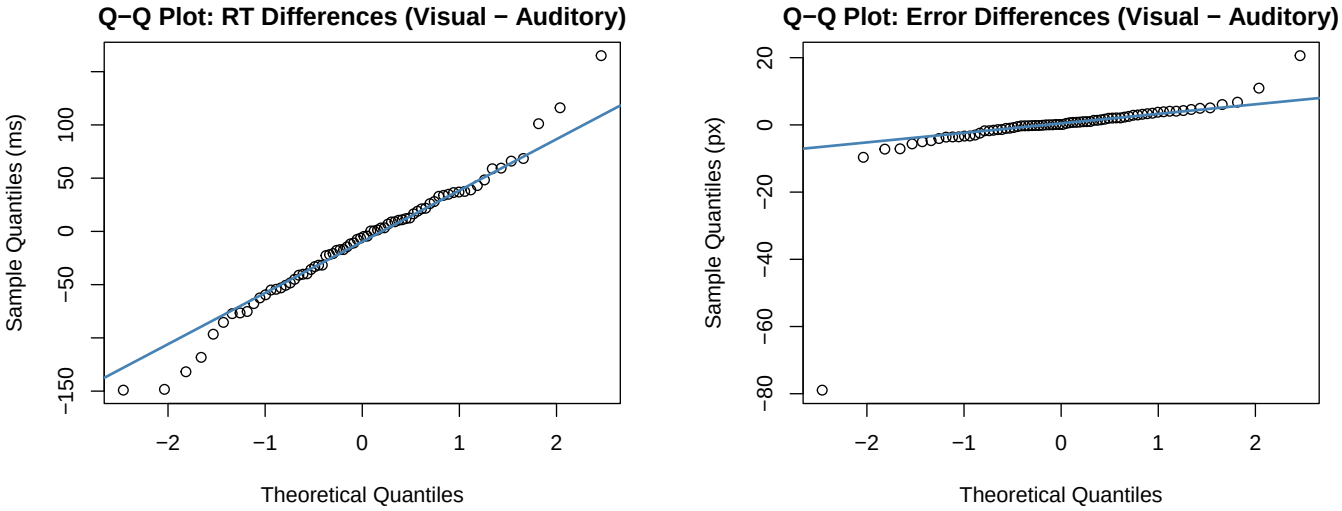
[6] R. P. Heitz, “The speed-accuracy tradeoff: history, physiology, methodology, and behavior,” *Front. Neurosci.*, 8, 150, 2014.

[7] M. Akamatsu et al., “Tactile, auditory, and visual feedback in a pointing task,” *Ergonomics*, 38(4), 816–827, 1995.

[8] R. Sigrist et al., “Augmented visual, auditory, haptic, and multimodal feedback in motor learning,” *Psychon. Bull. Rev.*, 20(1), 21–53, 2013.

A Assumption Checks

- RT differences: $W = 0.980$, $p = .304$: normal. Paired t-test appropriate.
- Error differences: $W = 0.420$, $p < .001$: non-normal. Wilcoxon used.



(a) RT differences:approximately normal. **(b)** Error differences:heavy tails, non-normal.

Figure 6: QQ plots for within-subject differences (visual − auditory).

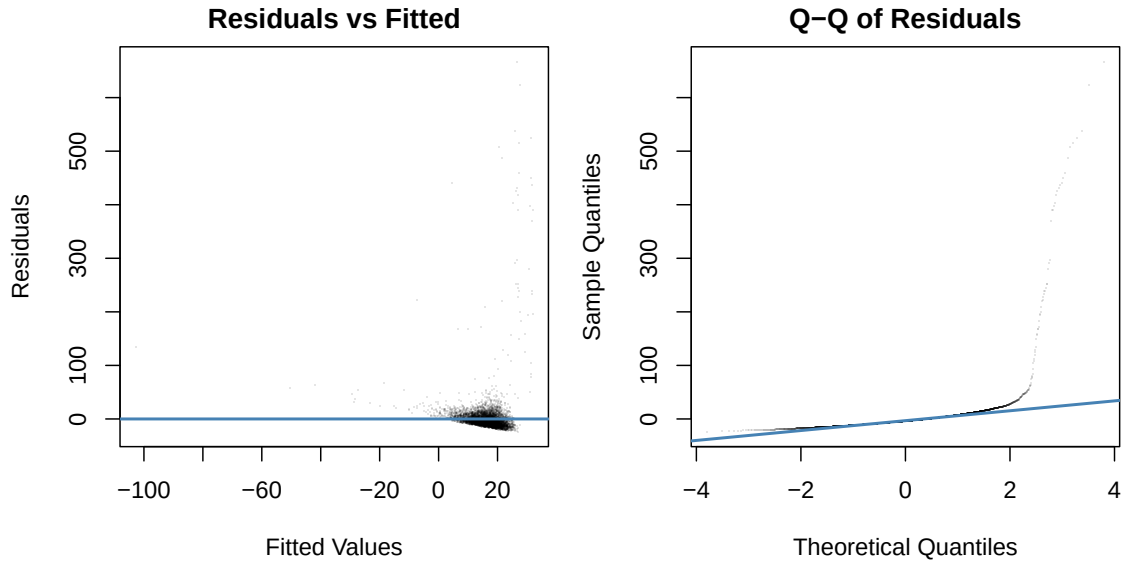
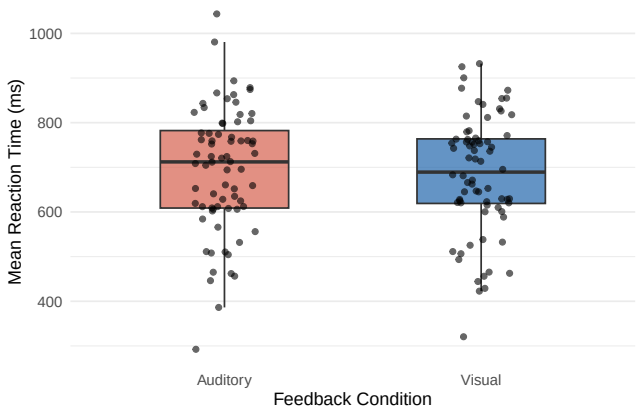
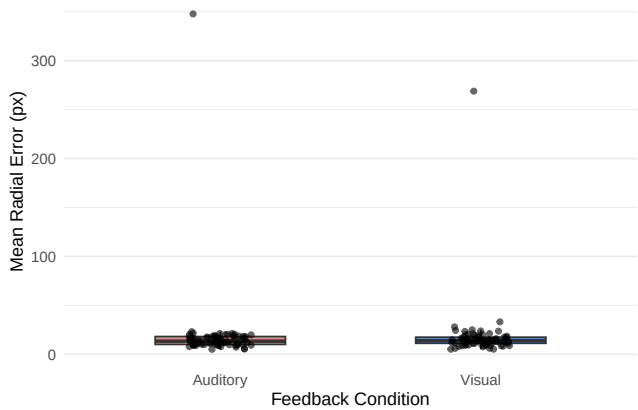


Figure 7: Residual diagnostics for the SAT interaction model. Funnel pattern expected with bounded error; heavy right tail from large misses.

B Additional Figures

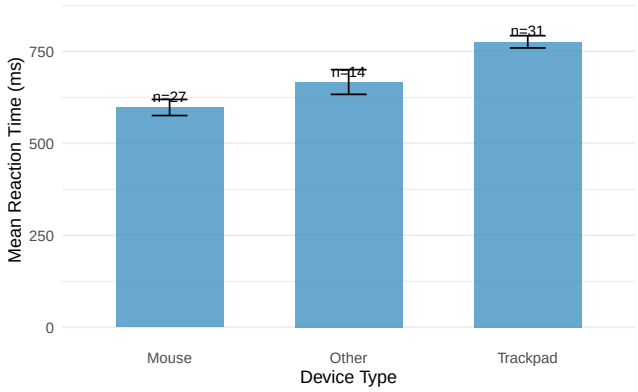


(a) RT by condition with individual points.

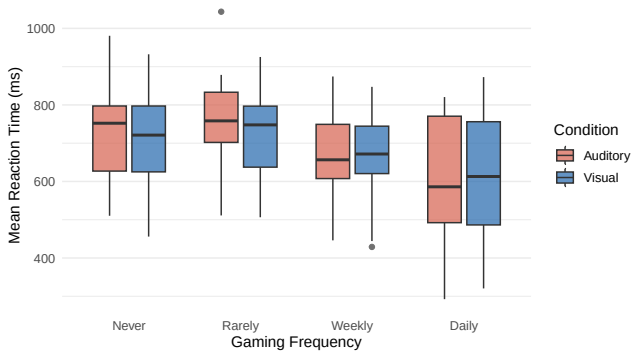


(b) Radial error by condition.

Figure 8: Boxplots of per-participant means by feedback condition.

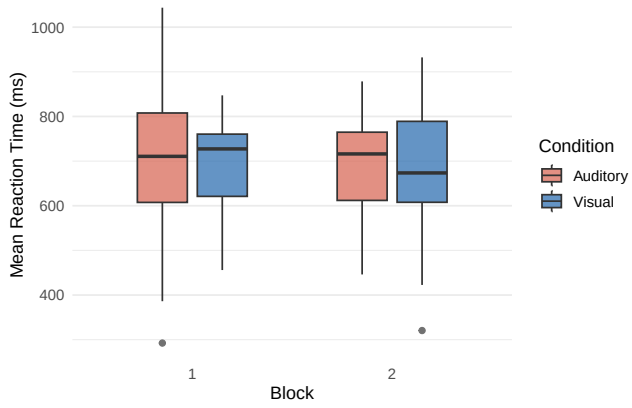


(a) RT by device type ($\pm$1 SE).

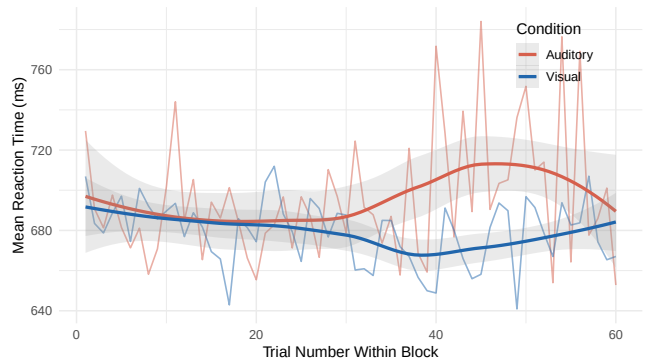


(b) RT by gaming frequency and condition.

Figure 9: Covariate effects on reaction time.

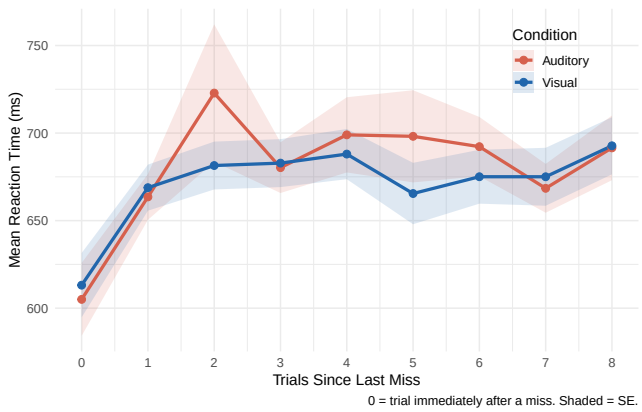


(a) RT by block ($p = .281$, $d = 0.13$).

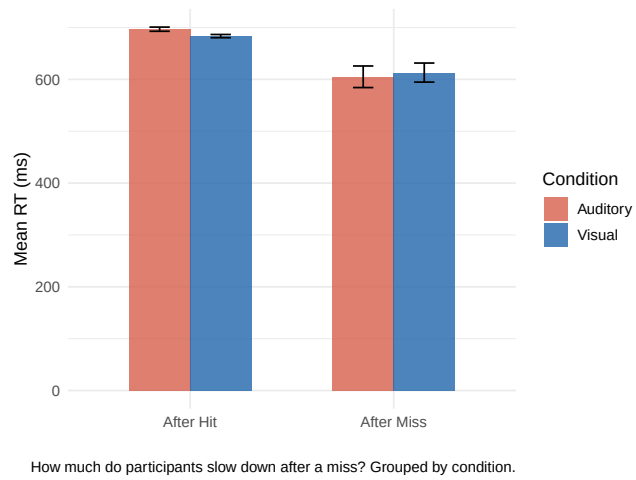


(b) RT by trial number within block.

Figure 10: Block and learning effects.



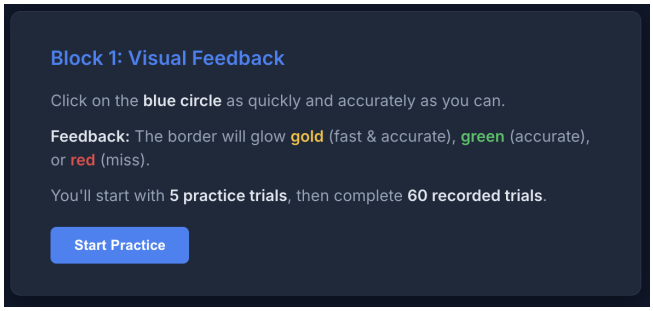
(a) RT trajectory after a miss. Visual participants slow down 65.5 ms vs. 17.4 ms for auditory.



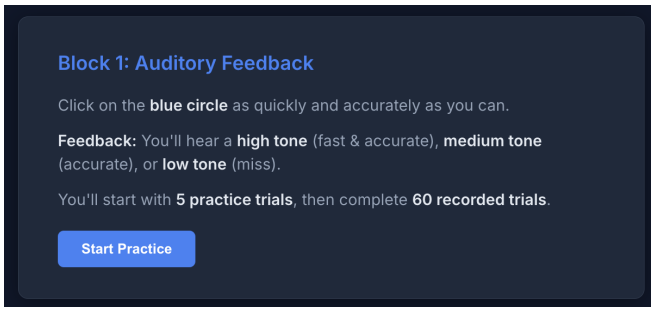
(b) Mean RT after hit vs. after miss, by condition.

Figure 11: Post-miss behavior: visual feedback makes failure more salient.

C Experiment Interface

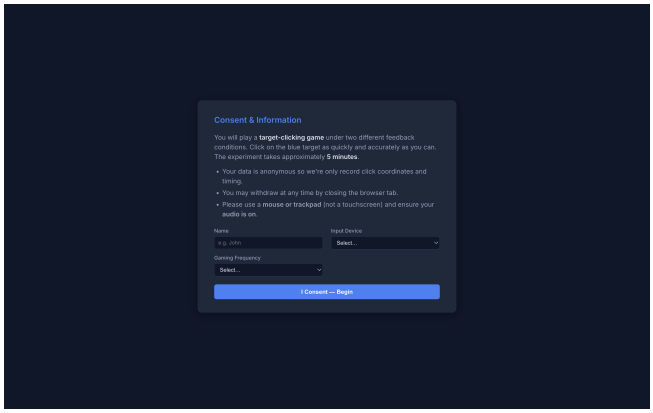


(a) Visual condition instructions.

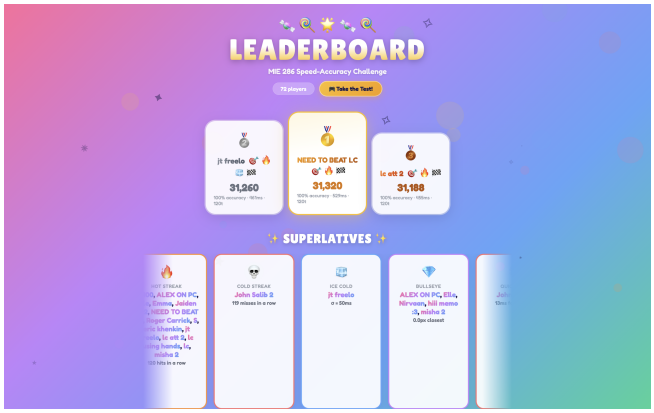


(b) Auditory condition instructions.

Figure 12: Instruction screens. Both describe the same task; only the feedback description differs.



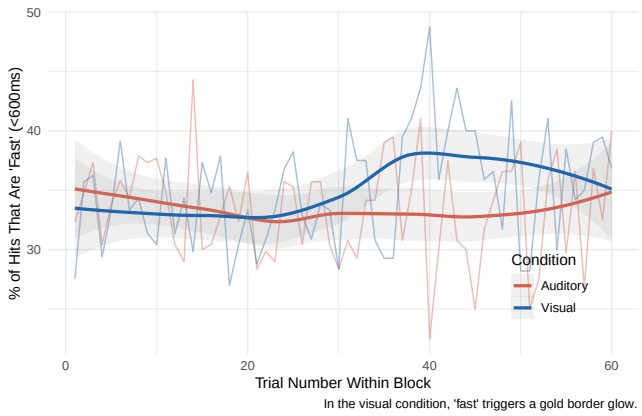
(a) Consent and demographics.



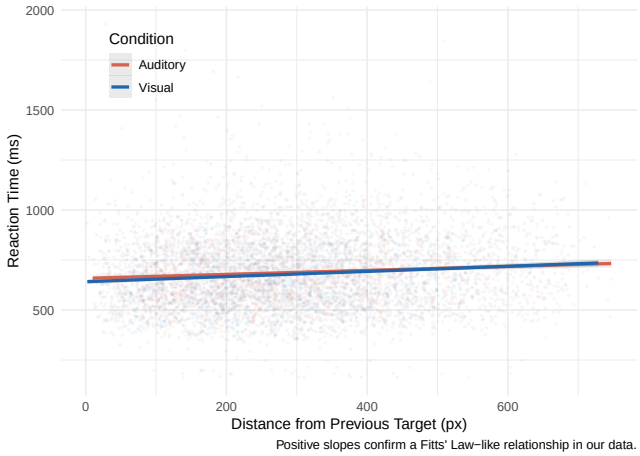
(b) Leaderboard with superlatives.

Figure 13: Data collection interface. The leaderboard was introduced mid-collection to boost engagement.

D Additional Analyses

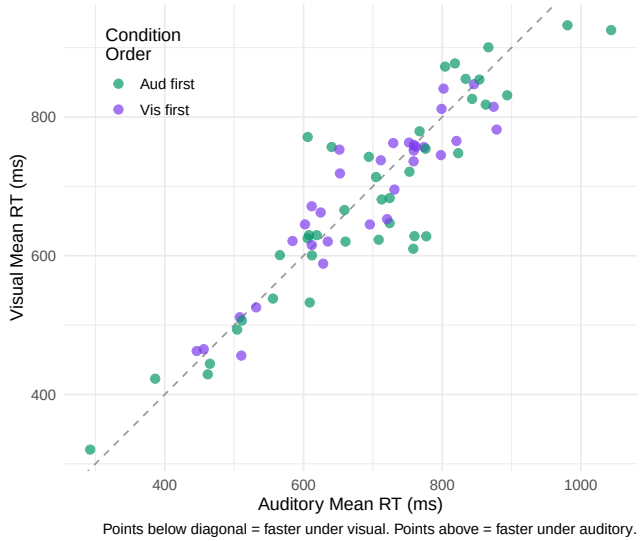


(a) Gold-chasing: % of fast ($<$600 ms) hits over trial number. Visual condition triggers gold glow for fast hits.

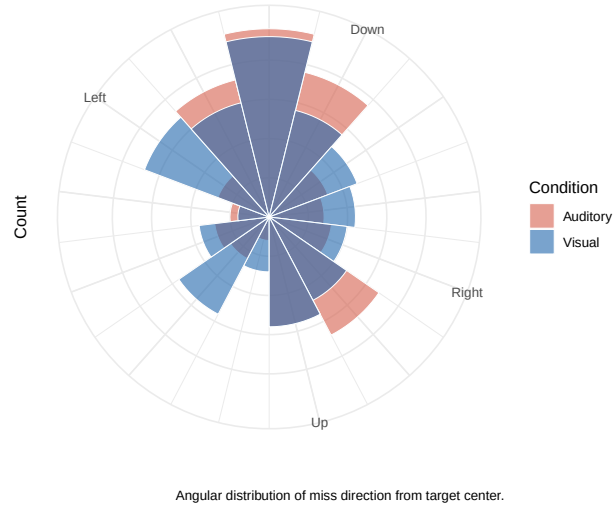


(b) Fitts' Law: inter-target distance predicts RT ($p < .001$). Visual slope is steeper (0.13 vs 0.10 ms/px).

Figure 14: Gold-chasing behavior and Fitts' Law relationships by condition.



(a) Per-participant: visual RT vs auditory RT. Below diagonal = faster under visual.



(b) Polar histogram of miss direction from target. Shows systematic rightward and downward bias.

Figure 15: Individual differences and spatial miss patterns.

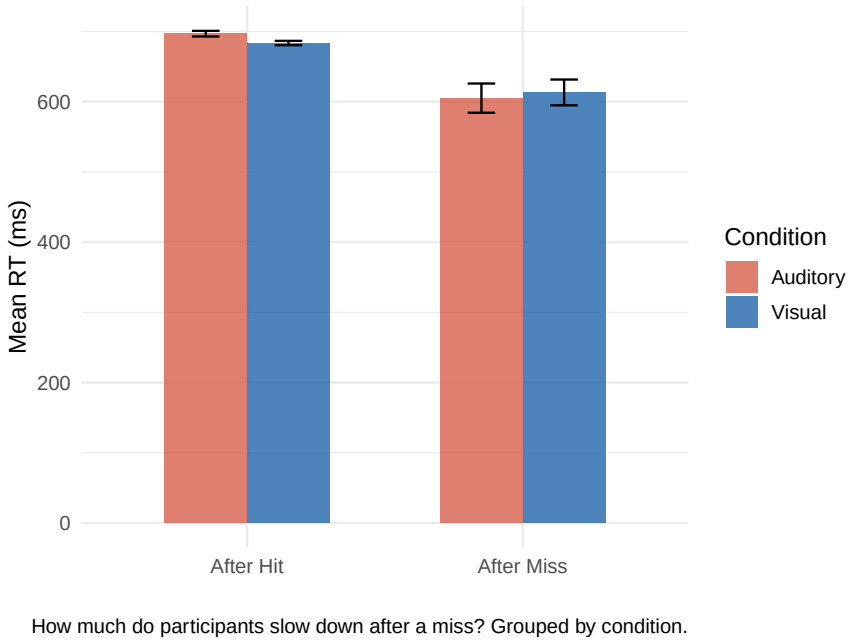


Figure 16: Mean RT after a hit vs. after a miss, by condition. Visual participants slow down 65.5 ms after a miss vs. 17.4 ms for auditory, suggesting visual feedback makes failure more salient.

E R Code

All analyses were conducted in R 4.5. The pipeline consists of six scripts run sequentially via `00_run_all.R`:

1. `01_load_clean.R` : Load 72 CSVs, exclude RT outliers, aggregate per-participant means.
2. `02_descriptives.R` : Summary tables by condition, device, gaming frequency.
3. `03_assumptions.R` : Shapiro–Wilk tests and QQ plots.
4. `04_inferential.R` : Paired t-test (H_1), Wilcoxon (H_2), multiple regression with interaction (H_3).
5. `05_plots.R` : All figures.
6. `06_secondary.R` : Tilt effect, order check, variance, block comparison, leaderboard effect, spatial bias.

Full code is in the `analysis/scripts/` directory submitted alongside this report.

F Attribution

Member	Contributions
Markiyan Konyk	
Jessica Yi	
Memo Ozdincer	